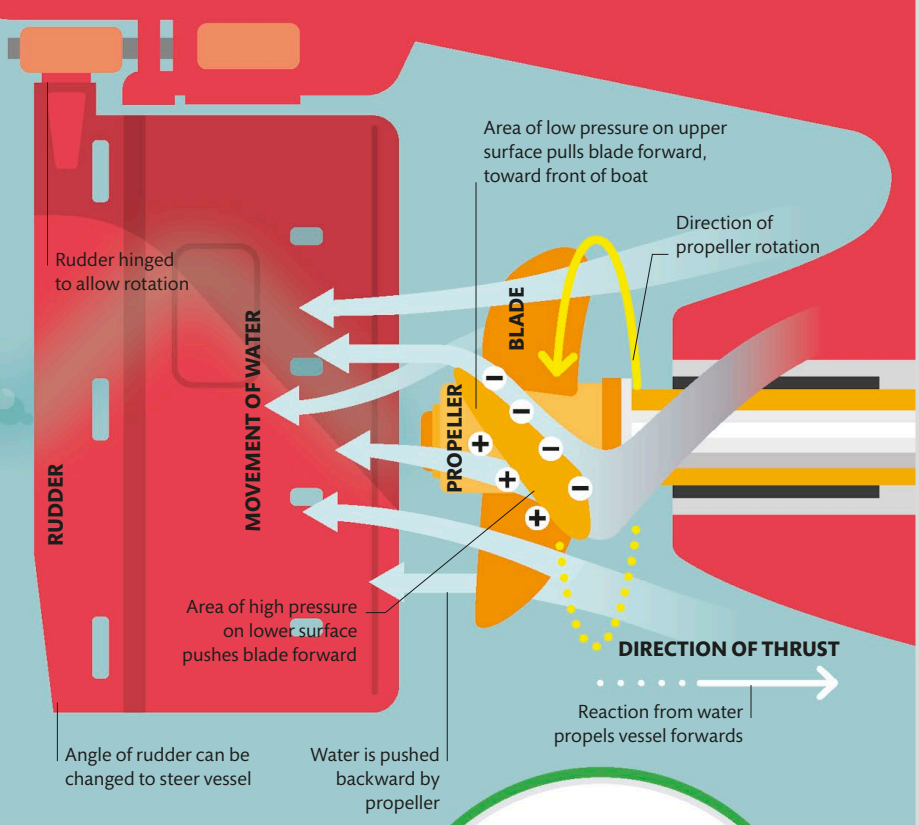
Propellers

A motor vessel's engine power is usually converted into the motion of the vessel through water by one or more propellers. When a propeller spins, its angled blades force water backward. The water pushes back against the propeller blades, generating thrust, which moves the boat. Water rushes in to fill the space that has been created behind the moving blade. This creates a pressure difference on either side of the blade, with low pressure at the front of the blade and high pressure behind. This pulls the front surface of the blade forward. Propellers are also called screws, because of their screwlike motion through water.



Motor vessels

The power provided by an engine frees motor vessels from the limitations of wind and sail. It also enables boats to generate electrical and hydraulic power to operate additional equipment.

WHAT IS THE FASTEST MOTORBOAT?

In 1978, Australian motorboat racer Ken Warby set a speed record of 317 miles (511 km) per hour in his jet-propelled powerboat.

Engines

A motor vessel can be powered in a number of different ways. Many use a diesel engine (see pp.42–43) to turn a shaft connected to a propeller. Other ships, including ocean liners, are powered by steam turbines. Warships often have gas-turbine engines similar to jet engines (see pp.60–61), and a handful of the biggest ships are nuclear-powered. On smaller boats, the engine is often mounted on the outside of the boat, while larger vessels usually have inboard motors.

Stability

A motor vessel's inboard engine may be used to power more than one propeller, as well as bow thrusters to help with steering (see opposite). The engine and heavy equipment are positioned low down in the hull to improve stability.

